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rechecked by Moeez
searchable by Khalil_Sahir

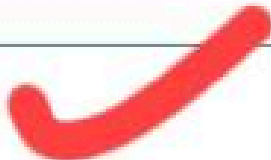
Question # 5 of 10 (Start time: 04:47:12 PM, 26 May 2021)

Total Marks: 1

The point to the left of which the area is 35% is known as

Select the correct option

- | | |
|-----------------------|-----------------|
| ☐ | 35th percentile |
| ☐ | Quartile |
| ☐ | 65/35 rule |
| ☐ | 35th value |



Question # 4 of 10 (Start time: 04:45:55 PM, 26 May 2021)

Total Marks: 1

Covariance is denoted by

Select the correct option

 $C(X,Y)$ 

CVR

 $Cov(X,Y)$ 

CV

Question # 3 of 10 (Start time: 04:33:18 PM, 26 May 2021)

Total Marks: 10

If x and y are two real numbers then what is the relationship between the absolute value (the modulus) of $|x+y|$ and the modulus of $|x|$ and modulus of $|y|$?

Select the correct option

[Reload Math Equation](#)

☐	$ x + y \leq x + y $
☐	$ x + y \neq x + y $
☐	$ x + y \geq x + y $
☐	$ x + y = x + y $

[Click to Save Answer & Move to Next Question](#)

Let X be a random variable with cumulative distribution function $F(x)$.
If $\lim_{x \rightarrow -\infty}$ then lower limit of $F(x)$ is

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|------------------|
| ☐ | $F(x) = 1$ |
| ☐ | $F(x) = -\infty$ |
| ☐ | $F(x) = -1$ |
| ☐ | $F(x) = 0$ |
- 

[Click to Save Answer & Move to Next Question](#)



MC200200242: AMNA ANWAR

Time Left 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 6 of 10 (Start time: 04:23:37 PM, 26 May 2021)

Total Marks: 1

In a continuous case, what will be the value of CDF for all those x values that lies between 0 and 1?

Select the correct option

☐ $F(x) = 0$

☐ $F(x) = 1$

☒ $F(x) = x$

☐ $F(x) = 0.5$

Click to Save Answer & Move to Next Question





MC200200242: AMNA ANWAR

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 2 of 10 (Start time: 04:17:12 PM, 26 May 2021)

Total Marks: 1

As the random variables, X_1 and X_2 are continuous random variables, the difference of these random variables is

Select the correct option

- ☐ discrete random variable
- ☒ continuous random variable
- ☐ both continuous and discrete
- ☐ not existent

Click to Save Answer & Move to Next Question





MC200200242: AMNA ANWAR

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 3 of 10 (Start time: 04:19:31 PM, 26 May 2021)

Total Marks: 1

In Probability, CDF stands for

Select the correct option

- ☐ Central Distribution Function
- ☐ Complete Distribution Function
- ☐ Cumulative Deviation Function
- ☒ Cumulative Distribution Function

Click to Save Answer & Move to Next Question





MC200200839: MUHAMMAD TARIQ

Time Left: 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 10 of 10 (Start time: 04:08:27 PM, 20 May 2021)

Total Marks: 1

What is the name of the following series?

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Select the correct option

[Reload Math Equations](#)

- | | |
|----------------------------------|--------------------|
| ☒ | Maclaurin series |
| ☐ | Exponential series |
| ☐ | Binomial series |
| ☐ | Taylor series |

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MC200200839: MUHAMMAD TARIQ

Time Left 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 8 of 10 (Start time: 04:05:39 PM, 20 May 2021)

Total Marks: 1

In probability, PDF stands for

Select the correct option



Probability Density Function



Probability Deviation Function



Probability Distribution Function



Probability Discrete Function

Click to Stop Answer & Move to Next Question



MC200200839: MUHAMMAD TARIQ

Time Left: 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 7 of 10 (Start time: 04:04:47 PM, 20 May 2021)

Total Marks: 1

If random variable X is infinite then its expectation

Select the correct option



minimum



exist



does not exist



is finite



Click to Stop Answer & Move to Next Question



MC200200839: MUHAMMAD TARIQ

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 4 of 10 (Start time: 04:01:34 PM, 20 May 2021)

Total Marks: 1

Joint probability distribution of two variables is called

Select the correct option



Binomial distribution



Multivariate distribution



Bivariate distribution



Normal Distribution

Circle the correct answer & Move to Next Question





MC200200839: MUHAMMAD TARIQ

Time Left: 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 3 of 10 (Start time: 04:00:27 PM, 20 May 2021)

Total Marks: 1

In a continuous case, what will be the value of CDF for all x values which are less than zero?

Select the correct option

☐ $F(x) = 0$

☐ $F(x) = x$

☐ $F(x) = -1$

☐ $F(x) = 1$

Click to Show Answer & Move to Next Question

MC200204868: MUHAMMAD RIZWAN

STA642 QUIZ - 1

Question # 8 of 10 (Start time: 03:48:37 PM, 26 May 2021)

The correlation coefficient can be expressed as

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | $\frac{\sigma_1 \sigma_2}{\text{cov}(X, Y)}$ |
| ☐ | $\frac{\text{cov}(X, Y)}{\bar{X}_1 \bar{X}_2}$ |
| ☒ | $\frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}$ |
| ☐ | $\frac{\text{cov}(X, Y)}{X_1 X_2}$ |

MC200204868: MUHAMMAD RIZWAN

STA642: QUIZ - 1

Question # 7 of 10 (Start time: 03:47:12 PM, 26 May 2021)

$$E(X_2) = E[E(X_2|X_1)]$$

The above formula is sometimes known as :

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☐ | law of double expectation |
| ☒ | law of iterated expectations |
| ☐ | law of variance |
| ☐ | law of second expectation |

Quiz List x Quiz x

← → ↻ 🔒 quiz.vu.edu.pk/QuizQuestion.aspx?ver=08f6932e-24a2-4

MC200204868: MUHAMMAD RIZWAN

STA642:QUIZ - 1

Question # 5 of 10 (Start time: 03:45:14 PM, 26 May 2021)

K(t) is equal to

Select the correct option

☐	$\log M(t)$
☐	$M(t)$
☐	$E(M(t))$
☐	$\text{Var}(M(t))$



MC200204868: MUHAMMAD RIZWAN

STA642: QUIZ - 1

Question # 4 of 10 (Start time: 03:43:57 PM, 26 May 2021)

The collection of all possible outcomes from an experiment is called

Select the correct option

- | | |
|-----------------------|--------------|
| ☐ | Sample frame |
| ☐ | Probability |
| ☐ | Event |
| ☐ | Sample Space |
- 

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MC200200723: LUBNA NAHEED

Time Left 5:11

Quiz Start Time: 04:55 PM, 29 May 2021

STAA42 QUIZ - 1

Total Marks: 1

Question # 7 of 10 (Start Time: 04:57:41 PM, 29 May 2021)

Generally, Probability density function is symbolically denoted by

Select the correct option

☐ $f(x)$ ☐ $d(x)$ ☒ $F(x)$ ☐ $L(x)$

Click to Save Answer & Move to Next Question



quiz.vu.edu.pk/QuizQuestio

1



MC200200723: LUBNA NAHEED

Time Left 5:4

STAA42 QUIZ - 1

Quiz Start Time: 04:55 PM, 29 May 2021



Question # 3 of 10 (Start time: 02:13:01 PM, 26 May 2021)

The concept of the CDF, will always be _____

Select the correct option



monotonically decreasing function



monotonically increasing function



monotonically zig-zag function



none of these

Click



00203588: MUHAMMAD SOHAIL

42:QUIZ - 1

tion # 6 of 10 (Start time: 02:17:14 PM, 26 May 2021)

The expected value of e^{itX} is called

at the correct option

☒	Characteristic Function
☐	Cumulative Distribution Function
☐	Probability Density Function
☐	Moment Generating Function



Question # 7 of 10 (Start time: 02:19:05 PM, 26 May 2021)

Toss a fair die; the possible values are 1, 2, 3, 4, 5, 6. Then probability is

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | $P(X = x) = \frac{3}{6}$ |
| ☐ | $P(X = x) = \frac{1}{1 + 2 + 3 + 4 + 5 + 6}$ |
| ☒ | $P(X = x) = \frac{1}{6}$ |
| ☐ | $P(X = x) = \frac{6}{6}$ |



Question # 8 of 10 (Start time: 02:20:09 PM, 26 May 2021)

$F(x)$ is often called simply the

Select the correct option



frequency function



negative function



distribution function



probability function



MUHAMMAD SOHAIL

Time Left

Quiz Start Time: 02:08 PM

(Start time: 02:23:01 PM, 26 May 2021)

In the continuous case of joint pdf $f_{X_1, X_2}(x_1, x_2)$, the marginal pdf of X_2 is found by integrating

option

Reload M

Neither X_1 nor X_2 X_2 X_1 Both X_1 and X_2

Click to Save Answer & Move to Next Question

Question 1 of 10 (Start time: 02:25:21 PM, 26 May 2021)

The point to the left of which the area is 20% is known as

Select the correct option

☐

20th value

☐

80/20 rule

☐

Quartile

☒

20th percentile



Question # 3 of 10 (Start time: 02:27:13 PM, 26 May 2021)

What will be the value of

Total Marks: 1

$$F_{XY}(\infty, \infty)$$

Select the correct option

Reload Math Equations



$$F_{XY}(\infty, \infty) = 1$$



$$F_{XY}(\infty, \infty) = 0$$



$$F_{XY}(\infty, \infty) = \infty$$



$$F_{XY}(\infty, \infty) = -\infty$$

Question # 6 of 10 (**Start time: 02:30:31 PM, 26 May 2021**)

The probability of an impossible event is

Select the correct option

☐ infinity

☐ $1/2$

☐ 1

☒ 0





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quiz.vu.edu.pk/QuizQuest

2



, 26 May 2021)

For all pdfs, the value of $F_x(\infty)$

1



- ∞

+ ∞

0



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Question # 7 of 10 (Start time: 02:31:33 PM, 26 May 2021)

Quiz Start Time: 02:25 PM, 26 May 2021

The sum of the first n natural numbers

Total Marks: 1

Select the correct option

 Reload Math Equations



$$n(n+2)$$



$$n(n+1)$$



$$\frac{n(n+2)}{2}$$



$$\frac{n(n+1)}{2}$$



Click to expand or collapse the question

Question # 8 of 10 (Start time: 02:32:32 PM, 26 May 2021)

Median is possible for which distribution?

Total Marks: 1

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☐ | Continuous distribution only |
| ☐ | None of the given options |
| ☒ | Both Discrete and Continuous |
| ☐ | Discrete distribution only |



The symbol $\varphi_x(t)$ is used to denote the

Select the correct option

 Reload Math Equations



Characteristic Function



Moment Generating Function



Cumulative Distribution Function



Probability Density Function

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

$$p(1) = \frac{12}{256}$$

$$p(1) = \frac{54}{256}$$

$$p(1) = \frac{108}{256}$$



$$p(1) = \frac{81}{256}$$



Question # 6 of 10 (Start time: 02:53:44 PM, 26 May 2021)

If random variable X is finite then its expectation

Select the correct option

- ☐ is finite
- ☒ exist 
- ☐ does not exist
- ☐ is infinite



Search the web and Windows



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f1

f2

f3



f4



f5

f6



f7

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MC200203376: MAAZ

Time Left 88 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 08:20 PM, 26 May 2021

Question # 4 of 10 (Start time: 08:23:40 PM, 26 May 2021)

Total Marks: 1

For a data we find correlation coefficient. Now, we added a digit 5 in all the data values and then again computed the correlation. The new correlation coefficient

_____.

Select the correct option



will decrease



will increase



will remain same



will increase by 5%



The conditional probability of event A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 1$$

$$P(B) \neq 0$$



$$P(B) = 0$$

$$P(B) = 1$$



f5

f6

f7

f8

f9

f10

f11

f12

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9

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For any distribution, if a higher moment exists, then _____

Select the correct option



all higher moments necessarily exists.



very next higher moments necessarily exists.



all lower moments necessarily exists



all lower moments may not exists



Search the web and Windows



esc

f1

f2

f3



f4



f5

f6



~

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%

5

Question # 9 of 10 (Start time: 02:57:15 PM, 26 May 2021)

Any experiment or process in which the outcome is _____ that is eligible to be called a random experiment.

Select the correct option

☐ Fixed

☐ Predictable

☐ Certain

☐ Unpredictable



Search the web and Windows





Quiz

<https://quiz.vu.edu.pk>



MC200203376: MAAZ

Time Left 88 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 08:20 PM, 26 May 2021

Question # 5 of 10 (Start time: 08:25:32 PM, 26 May 2021)

Total Marks: 1

If we divide or multiply all the values of X by any constant and divide or multiply all the values of Y by any constant, it will not affect the coefficient of correlation. This property is known as:

Select the correct option

- ☒ Coefficients of Correlation are dependent of Change of Scale
- ☐ Coefficients of Correlation are dependent of Change of Origin
- ☐ Coefficients of Correlation are independent of Change of Scale
- ☐ Coefficients of Correlation are independent of Change of Origin

Click on Save Answer & Move to Next Question



MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 1 of 10 (Start time: 09:16:56 PM, 26 May 2021)

If two variables X and Y are independent, coefficient of correlation between them will be

Select the correct option

- | | |
|----------------------------------|-----|
| ☒ | 0 |
| ☐ | 100 |
| ☐ | 1 |
| ☐ | 2 |

Click to Save Answer & Move to Next Question





MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 3 of 10 (Start time: 09:19:18 PM, 26 May 2021)

The point to the left of which the area is 20% is known as

Select the correct option

- | | |
|-----------------------|-----------------|
| ☐ | 80/20 rule |
| ☐ | 20th percentile |
| ☐ | 20th value |
| ☐ | Quartile |

Click to Save Answer & Move to Next Question





200203745: AYYAZ SIDDIQUE

642:QUIZ - 1

Question # 4 of 10 (Start time: 09:20:07 PM, 26 May 2021)

Consider the following formula.

$$\frac{E[(X - \mu_1)(X - \mu_2)]}{\sigma_1 \sigma_2}$$

Select the correct option

☒	Correlation
☐	Covariance
☐	Joint Variance
☐	Joint Mean

Click to Save Answer & Move to Next Question





STA642:QUIZ - 1

Question # 5 of 10 (Start time: 09:20:55 PM, 26 May 2021)

The probability of a sure event is

Select the correct option



1



0



infinity

 $1/2$

Click to Save Answer & Move to Next Question





MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 6 of 10 (Start time: 09:22:06 PM, 26 May 2021)

The point to the left of which the area is 70% is known as

Select the correct option

- | | |
|-----------------------|-----------------|
| ☐ | 70th value |
| ☐ | Quartile |
| ☐ | 70th percentile |
| ☐ | 70/30 rule |



Click to Save Answer & Move to Next Question



**Question # 7 of 10 (Start time: 09:22:53 PM, 26 May 2021)**

Central moment generating function is denoted by

Select the correct option

- | | |
|----------------------------------|---------|
| ☒ | C(t) |
| ☐ | CM |
| ☐ | CentMom |
| ☐ | C |

Click to Save Answer & Move to Next Question





UE

Quiz Sta

:24:38 PM, 26 May 2021)

Let $X = (X_1, X_2)$ be a random vector.
Then the expected value of X exists if the expectation(s) of

X_1 and X_2 both exist



X_2 exist only

X_1 and X_2 does not matter

X_1 exist only

Click to Save Answer & Move to Next Question





MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 10 of 10 (Start time: 09:25:27 PM, 26 May 2021)

Coefficient of Correlation is independent of Change of Scale. Symbolically, this property is shown as:

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☐ | $\rho_{U-V} = \rho_{XY}$ |
| ☐ | $\rho_{YX} = \rho_{XY}$ |
| ☐ | $\rho_{U+V} = \rho_{XY}$ |
| ☒ | $\rho_{UV} = \rho_{XY}$ |



Click to Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 1 of 10 (Start time: 09:29:25 PM, 26 May 2021)

Total Marks: 1

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--------------------------|
| ☐ | $p(1) = \frac{108}{256}$ |
| ☐ | $p(1) = \frac{12}{256}$ |
| ☐ | $p(1) = \frac{54}{256}$ |
| ☐ | $p(1) = \frac{81}{256}$ |

Click on Save Answer & Next to Next Question





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 2 of 10 (Start time: 09:30:04 PM, 26 May 2021)

Total Marks: 1

The Cumulative distribution graph for the discrete case is like a

Select the correct option

☐ staircase☐ zig-zag☐ straight line☐ spiral

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642 QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 4 of 10 (Start time: 09:32:55 PM, 26 May 2021)

Total Marks: 1

The joint probability distribution can be expressed in terms of a

Select the correct option

- ☒ all of above three options
- ☐ joint cumulative distribution function
- ☐ joint probability mass function
- ☐ joint probability density function

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 5 of 10 (Start time: 09:34:22 PM, 26 May 2021)

Total Marks: 1

The quantile function is also called the

Select the correct option

☐ mod function☐ inverse cumulative distribution function☐ cumulative distribution function☐ median function

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 6 of 10 (Start time: 09:35:59 PM, 26 May 2021)

Total Marks: 1

The point to the left of which the area is 35% is known as

Select the correct option

- | | |
|-----------------------|-----------------|
| ☐ | 35th value |
| ☐ | 35th percentile |
| ☐ | Quartile |
| ☐ | 65/35 rule |



Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 86 sec(s)

STA642 QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 7 of 10 (Start time: 09:36:42 PM, 26 May 2021)

Total Marks: 1

Jacobian transformation is denoted by

Select the correct option

- | | |
|----------------------------------|------|
| ☐ | J(t) |
| ☐ | JCob |
| ☒ | J |
| ☐ | Jt |

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 85 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 9 of 10 (Start time: 09:40:22 PM, 26 May 2021)

Total Marks: 1

For a function $x = \frac{1}{2\sqrt{y}}$, what will be the value of x for $y = 0.6$

Select the correct option

Reload Math Equations

☐	0.42
☐	1.65
☒	0.65
☐	0.2

Click to Load Answer & Check My Test Answers





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 10 of 10 (Start time: 09:41:14 PM, 26 May 2021)

Total Marks: 1

If we subtract or add any constant in ALL values of X and Y, it will not affect the coefficient of correlation. This property is known as:

Select the correct option

- ☐ Coefficients of Correlation are dependent of Change of Origin
- ☐ Coefficients of Correlation are dependent of Change of Scale
- ☐ Coefficients of Correlation are independent of Change of Scale
- ☐ Coefficients of Correlation are independent of Change of Origin

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 3 of 10 (Start time: 09:30:59 PM, 26 May 2021)

Total Marks: 1

Theoretically there can be _____ of decimal places after the decimals.

Select the correct option

☐ two number☒ an infinite number☐ fixed number☐ finite number

Click on Save Answer & Move to Next Question



Question # 5 of 10 (Start time: 04:47:12 PM, 26 May 2021)

Total Marks: 1

The point to the left of which the area is 35% is known as

Select the correct option

☒	35th percentile
☐	Quartile
☐	65/35 rule
☐	35th value

Question # 4 of 10 (Start time: 04:45:55 PM, 26 May 2021)

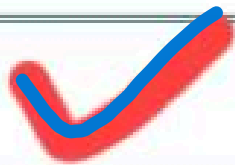
Total Marks: 1

Covariance is denoted by

Select the correct option

 $C(X,Y)$ 

CVR

 $Cov(X,Y)$ 

CV

Question # 3 of 10 (Start time: 04:33:18 PM, 26 May 2021)

Total Marks: 10

If x and y are two real numbers then what is the relationship between the absolute value (the modulus) of $|x+y|$ and the modulus of $|x|$ and modulus of $|y|$?

Select the correct option

[Reload Math Equation](#)

☐	$ x + y \leq x + y $
☐	$ x + y \neq x + y $
☐	$ x + y \geq x + y $
☐	$ x + y = x + y $

[Click to Save Answer & Move to Next Question](#)

Let X be a random variable with cumulative distribution function $F(x)$.
If $\lim_{x \rightarrow -\infty}$ then lower limit of $F(x)$ is

Select the correct option

 Reload Math Equations

☐	$F(x) = 1$
☐	$F(x) = -\infty$
☐	$F(x) = -1$
☐	$F(x) = 0$



Click to Save Answer & Move to Next Question



MC200200242: AMNA ANWAR

Time Left 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 6 of 10 (Start time: 04:23:37 PM, 26 May 2021)

Total Marks: 1

In a continuous case, what will be the value of CDF for all those x values that lies between 0 and 1?

Select the correct option

☐ $F(x) = 0$ ☐ $F(x) = 1$ ☒ $F(x) = x$ ☐ $F(x) = 0.5$

Click to Save Answer & Move to Next Question





MC200200242: AMNA ANWAR

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

page no 62

Question # 2 of 10 (Start time: 04:17:12 PM, 26 May 2021)

Total Marks: 1

As the random variables, X_1 and X_2 are continuous random variables, the difference of these random variables is

Select the correct option

- ☐ discrete random variable
- ☒ continuous random variable
- ☐ both continuous and discrete
- ☐ not existent

Click to Save Answer & Move to Next Question





MC200200242: AMNA ANWAR

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 3 of 10 (Start time: 04:19:31 PM, 26 May 2021)

Total Marks: 1

In Probability, CDF stands for

Select the correct option

- ☐ Central Distribution Function
- ☐ Complete Distribution Function
- ☐ Cumulative Deviation Function
- ☒ Cumulative Distribution Function

Click to Save Answer & Move to Next Question





MC200200839: MUHAMMAD TARIQ

Time Left: 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 10 of 10 (Start time: 04:08:27 PM, 20 May 2021)

Total Marks: 1

What is the name of the following series?

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Select the correct option

Reload Math Equations

☒	Maclaurin series
☐	Exponential series
☐	Binomial series
☐	Taylor series

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MC200200839: MUHAMMAD TARIQ

Time Left 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 8 of 10 (Start time: 04:05:39 PM, 20 May 2021)

Total Marks: 1

In probability, PDF stands for

Select the correct option



Probability Density Function



Probability Deviation Function



Probability Distribution Function



Probability Discrete Function

Click to Stop Answer & Move to Next Question





MC200200839: MUHAMMAD TARIQ

Time Left: 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 7 of 10 (Start time: 04:04:47 PM, 20 May 2021)

Total Marks: 1

If random variable X is infinite then its expectation

Select the correct option



minimum



exist



does not exist



is finite



Click to Stop Answer & Move to Next Question



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MC200200839: MUHAMMAD TARIQ

Time Left 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 4 of 10 (Start time: 04:01:34 PM, 20 May 2021)

Total Marks: 1

Joint probability distribution of two variables is called

Select the correct option



Binomial distribution



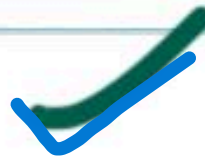
Multivariate distribution



Bivariate distribution



Normal Distribution



Circle to Stop Answer & Move to Next Question





MC200200839: MUHAMMAD TARIQ

Time Left: 86 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 20 May 2021

Question # 3 of 10 (Start time: 04:00:27 PM, 20 May 2021)

Total Marks: 1

In a continuous case, what will be the value of CDF for all x values which are less than zero?

Select the correct option



$$F(x) = 0$$



$$F(x) = x$$



$$F(x) = -1$$



$$F(x) = 1$$

Click to Show Answer & Move to Next Question

MC200204868: MUHAMMAD RIZWAN

STA642 QUIZ - 1

Question # 8 of 10 (Start time: 03:48:37 PM, 26 May 2021)

The correlation coefficient can be expressed as:

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | $\frac{\sigma_1 \sigma_2}{\text{cov}(X, Y)}$ |
| ☐ | $\frac{\text{cov}(X, Y)}{\bar{X}_1 \bar{X}_2}$ |
| ☒ | $\frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}$ |
| ☐ | $\frac{\text{cov}(X, Y)}{\bar{X}_1 \bar{X}_2}$ |

page no 76

MC200204868: MUHAMMAD RIZWAN

STA642: QUIZ - 1

Question # 7 of 10 (Start time: 03:47:12 PM, 26 May 2021)

$$E(X_2) = E[E(X_2|X_1)]$$

The above formula is sometimes known as :

Select the correct option

- | | |
|----------------------------------|------------------------------|
| ☐ | law of double expectation |
| ☒ | law of iterated expectations |
| ☐ | law of variance |
| ☐ | law of second expectation |

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MC200204868: MUHAMMAD RIZWAN

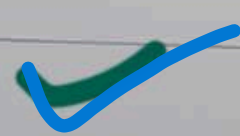
STA642:QUIZ - 1

Question # 5 of 10 (Start time: 03:45:14 PM, 26 May 2021)

K(t) is equal to

Select the correct option

☒	$\log M(t)$
☐	$M(t)$
☐	$E(M(t))$
☐	$\text{Var}(M(t))$



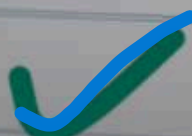
MC200204868: MUHAMMAD RIZWAN

STA642: QUIZ - 1

Question # 4 of 10 (Start time: 03:43:57 PM, 26 May 2021)

The collection of all possible outcomes from an experiment is called

Select the correct option

- | | |
|-----------------------|--------------|
| ☐ | Sample frame |
| ☐ | Probability |
| ☐ | Event |
| ☐ | Sample Space |
- 

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MC200200723: LUBNA NAHEED

Time Left 5:11

Quiz Start Time: 04:55 PM, 29 May 2021

STAA42 QUIZ - 1

Total Marks: 1

Question # 7 of 10 (Start Time: 04:57:41 PM, 29 May 2021)

Generally, Probability density function is symbolically denoted by

Select the correct option

☐ $f(x)$ ☐ $d(x)$ ☒ $F(x)$ ☐ $L(x)$

Click to Save Answer & Move to Next Question



quiz.vu.edu.pk/QuizQuestio

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MC200200723: LUBNA NAHEED

Time Left 5:4

Quiz Start Time: 04:55 PM, 29 May 2021

STAA42 QUIZ - 1

Question # 7 of 10



Question # 3 of 10 (Start time: 02:13:01 PM, 26 May 2021)

The concept of the CDF, will always be _____

Select the correct option



monotonically decreasing function



monotonically increasing function



monotonically zig-zag function



none of these

Click



00203588: MUHAMMAD SOHAIL

42:QUIZ - 1

tion # 6 of 10 (Start time: 02:17:14 PM, 26 May 2021)

The expected value of e^{itX} is called

at the correct option

☒	Characteristic Function
☐	Cumulative Distribution Function
☐	Probability Density Function
☐	Moment Generating Function



Question # 7 of 10 (Start time: 02:19:05 PM, 26 May 2021)

Toss a fair die; the possible values are 1, 2, 3, 4, 5, 6. Then probability is

Select the correct option

- | | |
|-----------------------|--|
| ☐ | $P(X = x) = \frac{3}{6}$ |
| ☐ | $P(X = x) = \frac{1}{1 + 2 + 3 + 4 + 5 + 6}$ |
| ☐ | $P(X = x) = \frac{1}{6}$ |
| ☐ | $P(X = x) = \frac{6}{6}$ |





Question # 8 of 10 (Start time: 02:20:09 PM, 26 May 2021)

$F(x)$ is often called simply the

Select the correct option



frequency function



negative function



distribution function



probability function



MUHAMMAD SOHAIL

Time Left

Quiz Start Time: 02:08 PM

(Start time: 02:23:01 PM, 26 May 2021)

In the continuous case of joint pdf $f_{X_1, X_2}(x_1, x_2)$, the marginal pdf of X_2 is found by integrating

option

Reload M

Neither X_1 nor X_2 X_2 X_1 Both X_1 and X_2

Click to Save Answer & Move to Next Question

Question 1 of 10 (Start time: 02:25:21 PM, 26 May 2021)

The point to the left of which the area is 20% is known as

Select the correct option

☐

20th value

☐

80/20 rule

☐

Quartile

☐

20th percentile



Question # 3 of 10 (Start time: 02:27:13 PM, 26 May 2021)

What will be the value of

Total Marks: 1

$$F_{XY}(\infty, \infty)$$

Select the correct option

Reload Math Equations



$$F_{XY}(\infty, \infty) = 1$$



$$F_{XY}(\infty, \infty) = 0$$



$$F_{XY}(\infty, \infty) = \infty$$



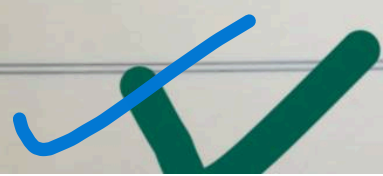
$$F_{XY}(\infty, \infty) = -\infty$$

Question # 6 of 10 (**Start time: 02:30:31 PM, 26 May 2021**)

The probability of an impossible event is

Select the correct option

- | | |
|----------------------------------|----------|
| ☐ | infinity |
| ☐ | $1/2$ |
| ☐ | 1 |
| ☒ | 0 |





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2



, 26 May 2021)

For all pdfs, the value of $F_x(\infty)$

1



- ∞

+ ∞

0



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Question # 7 of 10 (Start time: 02:31:33 PM, 26 May 2021)

Quiz Start Time: 02:25 PM, 26 May 2021

The sum of the first n natural numbers

Total Marks: 1

Select the correct option

 Reload Math Equations



$$n(n+2)$$



$$n(n+1)$$



$$\frac{n(n+2)}{2}$$



$$\frac{n(n+1)}{2}$$



Click to expand/collapse the question

Question # 8 of 10 (Start time: 02:32:32 PM, 26 May 2021)

Median is possible for which distribution?

Total Marks: 1

Select the correct option

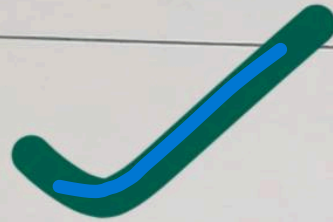
- | | |
|----------------------------------|------------------------------|
| ☐ | Continuous distribution only |
| ☐ | None of the given options |
| ☒ | Both Discrete and Continuous |
| ☐ | Discrete distribution only |



The symbol $\varphi_x(t)$ is used to denote the

Select the correct option

 Reload Math Equations



☐ Characteristic Function

☐ Moment Generating Function

☐ Cumulative Distribution Function

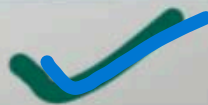
☐ Probability Density Function

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

$$p(1) = \frac{12}{256}$$

$$p(1) = \frac{54}{256}$$

$$p(1) = \frac{108}{256}$$



$$p(1) = \frac{81}{256}$$



Question # 6 of 10 (Start time: 02:53:44 PM, 26 May 2021)

If random variable X is finite then its expectation

Select the correct option

- ☐ is finite
- ☒ exist
- ☐ does not exist
- ☐ is infinite



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MC200203376: MAAZ

Time Left 88 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 08:20 PM, 26 May 2021

Question # 4 of 10 (Start time: 08:23:40 PM, 26 May 2021)

Total Marks: 1

For a data we find correlation coefficient. Now, we added a digit 5 in all the data values and then again computed the correlation. The new correlation coefficient

_____.

Select the correct option



will decrease



will increase



will remain same



will increase by 5%



The conditional probability of event A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 1$$

$$P(B) \neq 0$$

$$P(B) = 0$$

$$P(B) = 1$$

page no 73



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For any distribution, if a higher moment exists, then _____

Select the correct option

- ☐ all higher moments necessarily exists.
- ☐ very next higher moments necessarily exists.
- ☒ all lower moments necessarily exists
- ☐ all lower moments may not exists



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f4



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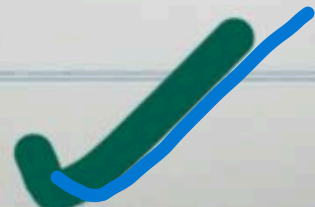
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Question # 9 of 10 (Start time: 02:57:15 PM, 26 May 2021)

Any experiment or process in which the outcome is _____ that is eligible to be called a random experiment.

Select the correct option

- | | |
|-----------------------|---------------|
| ☐ | Fixed |
| ☐ | Predictable |
| ☐ | Certain |
| ☐ | Unpredictable |
- 



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MC200203376: MAAZ

Time Left 88 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 08:20 PM, 26 May 2021

Question # 5 of 10 (Start time: 08:25:32 PM, 26 May 2021)

Total Marks: 1

If we divide or multiply all the values of X by any constant and divide or multiply all the values of Y by any constant, it will not affect the coefficient of correlation. This property is known as:

Select the correct option



Coefficients of Correlation are dependent of Change of Scale



Coefficients of Correlation are dependent of Change of Origin



Coefficients of Correlation are independent of Change of Scale



Coefficients of Correlation are independent of Change of Origin

Click on Save Answer & Move to Next Question

MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 1 of 10 (Start time: 09:16:56 PM, 26 May 2021)

If two variables X and Y are independent, coefficient of correlation between them will be

Select the correct option

- | | |
|----------------------------------|-----|
| ☒ | 0 |
| ☐ | 100 |
| ☐ | 1 |
| ☐ | 2 |

Click to Save Answer & Move to Next Question



MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 3 of 10 (Start time: 09:19:18 PM, 26 May 2021)

The point to the left of which the area is 20% is known as

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | 80/20 rule |
| ☒ | 20th percentile |
| ☐ | 20th value |
| ☐ | Quartile |

Click to Save Answer & Move to Next Question





200203745: AYYAZ SIDDIQUE

642:QUIZ - 1

Question # 4 of 10 (Start time: 09:20:07 PM, 26 May 2021)

Consider the following formula.

$$\frac{E[(X - \mu_1)(X - \mu_2)]}{\sigma_1 \sigma_2}$$

Select the correct option

☒	Correlation
☐	Covariance
☐	Joint Variance
☐	Joint Mean

Click to Save Answer & Move to Next Question





STA642:QUIZ - 1

Question # 5 of 10 (Start time: 09:20:55 PM, 26 May 2021)

The probability of a sure event is

Select the correct option



1



0



infinity

 $1/2$

Click to Save Answer & Move to Next Question





MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 6 of 10 (Start time: 09:22:06 PM, 26 May 2021)

The point to the left of which the area is 70% is known as

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | 70th value |
| ☐ | Quartile |
| ☒ | 70th percentile |
| ☐ | 70/30 rule |

Click to Save Answer & Move to Next Question



**Question # 7 of 10 (Start time: 09:22:53 PM, 26 May 2021)**

Central moment generating function is denoted by

Select the correct option

- | | |
|-----------------------|---------|
| ☐ | C(t) |
| ☐ | CM |
| ☐ | CentMom |
| ☐ | C |

Click to Save Answer & Move to Next Question





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Quiz Sta

:24:38 PM, 26 May 2021)

Let $X = (X_1, X_2)$ be a random vector.
Then the expected value of X exists if the expectation(s) of

X_1 and X_2 both exist

X_2 exist only

X_1 and X_2 does not matter

X_1 exist only

Click to Save Answer & Move to Next Question





MC200203745: AYYAZ SIDDIQUE

STA642:QUIZ - 1

Question # 10 of 10 (Start time: 09:25:27 PM, 26 May 2021)

Coefficient of Correlation is independent of Change of Scale. Symbolically, this property is shown as:

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☐ | $\rho_{U-V} = \rho_{XY}$ |
| ☐ | $\rho_{YX} = \rho_{XY}$ |
| ☐ | $\rho_{U+V} = \rho_{XY}$ |
| ☒ | $\rho_{UV} = \rho_{XY}$ |

Click to Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 1 of 10 (Start time: 09:29:25 PM, 26 May 2021)

Total Marks: 1

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

Select the correct option

Reload Math Equations

☐	$p(1) = \frac{108}{256}$
☐	$p(1) = \frac{12}{256}$
☐	$p(1) = \frac{54}{256}$
☐	$p(1) = \frac{81}{256}$

Click on Load Answer & Check My Test Answer





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 2 of 10 (Start time: 09:30:04 PM, 26 May 2021)

Total Marks: 1

The Cumulative distribution graph for the discrete case is like a

Select the correct option



staircase



zig-zag



straight line



spiral

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642 QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 4 of 10 (Start time: 09:32:55 PM, 26 May 2021)

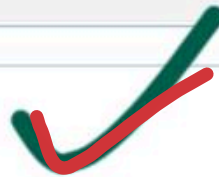
Total Marks: 1

The joint probability distribution can be expressed in terms of a

Select the correct option



all of above three options



joint cumulative distribution function



joint probability mass function



joint probability density function

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

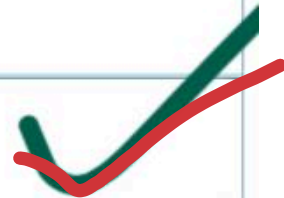
Quiz Start Time: 09:29 PM, 26 May 2021

Question # 5 of 10 (Start time: 09:34:22 PM, 26 May 2021)

Total Marks: 1

The quantile function is also called the

Select the correct option

☐ mod function☐ inverse cumulative distribution function☐ cumulative distribution function☐ median function

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 87 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 6 of 10 (Start time: 09:35:59 PM, 26 May 2021)

Total Marks: 1

The point to the left of which the area is 35% is known as

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | 35th value |
| ☒ | 35th percentile |
| ☐ | Quartile |
| ☐ | 65/35 rule |

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 86 sec(s)

STA642 QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 7 of 10 (Start time: 09:36:42 PM, 26 May 2021)

Total Marks: 1

Jacobian transformation is denoted by

Select the correct option

- | | |
|----------------------------------|------|
| ☐ | J(t) |
| ☐ | JCob |
| ☒ | J |
| ☐ | Jt |

page no 23

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 85 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 9 of 10 (Start time: 09:40:22 PM, 26 May 2021)

Total Marks: 1

For a function $x = \frac{1}{2\sqrt{y}}$, what will be the value of x for $y = 0.6$

Select the correct option

Reload Math Equations

☐	0.42
☐	1.65
☒	0.65
☐	0.2

Click to Load Answer & Check My Test Results





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 10 of 10 (Start time: 09:41:14 PM, 26 May 2021)

Total Marks: 1

If we subtract or add any constant in ALL values of X and Y, it will not affect the coefficient of correlation. This property is known as:

Select the correct option

- ☐ Coefficients of Correlation are dependent of Change of Origin
- ☐ Coefficients of Correlation are dependent of Change of Scale
- ☒ Coefficients of Correlation are independent of Change of Scale
- ☐ Coefficients of Correlation are independent of Change of Origin

Click on Save Answer & Move to Next Question





MC200201181: TANVEER ABBAS

Time Left 88 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 09:29 PM, 26 May 2021

Question # 3 of 10 (Start time: 09:30:59 PM, 26 May 2021)

Total Marks: 1

Theoretically there can be _____ of decimal places after the decimals.

Select the correct option

☐ two number☐ an infinite number☐ fixed number☐ finite number

Click on Save Answer & Move to Next Question





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MC200200829: MUHAMMAD TARIQ

Time Left 87 sec(s)

ETA642:QUIZ - 1

Quiz Start Time: 03:59 PM, 28 May 2021

Question # 10 of 10 (Start Time: 04:08:27 PM, 28 May 2021)

Total Marks:

What is the name of the following series?

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Select the correct option

Related Math Equations

☐	Maclaurin series
☐	Exponential series
☐	Binomial series
☐	Taylor series



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Question # 10 of 10 (Start time: 02:58:00 PM, 26 May 2021)

The probability of an impossible event is

Select the correct option



1



infinity



0



$\frac{1}{2}$

Question # 9 of 10 (Start time: 02:57:15 PM, 26 May 2021)

Any experiment or process in which the outcome is _____ that is eligible to be called a random experiment.

Select the correct option

- | | |
|-----------------------|---------------|
| ☐ | Fixed |
| ☐ | Predictable |
| ☐ | Certain |
| ☐ | Unpredictable |

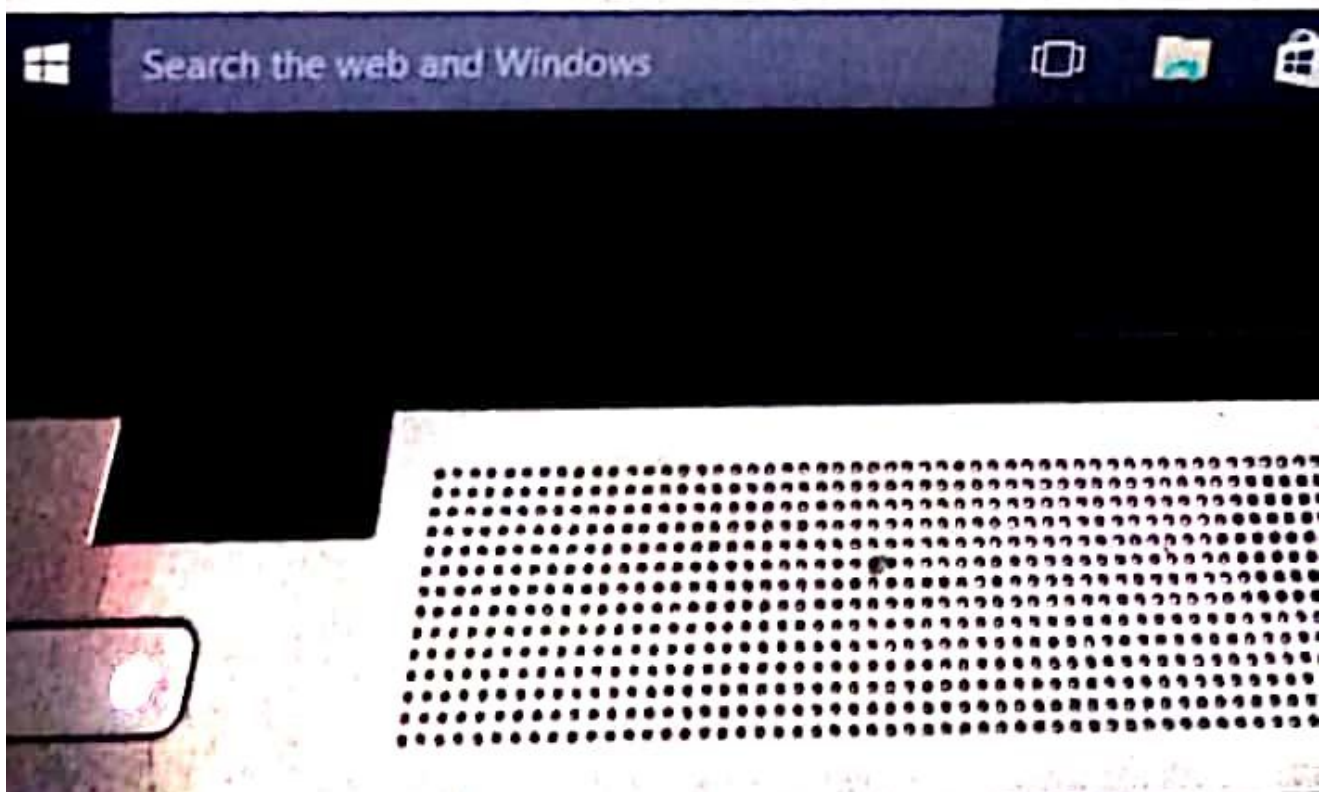
page no 1

For any distribution, if a higher moment exists, then _____

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | all higher moments necessarily exists |
| ☐ | very next higher moments necessarily exists |
| ☒ | all lower moments necessarily exists |
| ☐ | all lower moments may not exists |

page no 48



The conditional probability of event A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 1$$



$$P(B) \neq 0$$



page no 73

$$P(B) = 0$$

$$P(B) = 1$$

Question # 6 of 10 (Start time: 02:53:44 PM, 26 May 2021)

If random variable X is finite then its expectation

Select the correct option

- ☐ is finite
- ☒ exist
- ☐ does not exist
- ☐ is infinite



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For a function $x = \frac{1}{2\sqrt{y}}$, what will be the value of x for $y = 0.6$

 Select the correct option

- | | |
|----------------------------------|------|
| ☐ | 1.65 |
| ☒ | 0.65 |
| ☐ | 0.42 |
| ☐ | 0.2 |



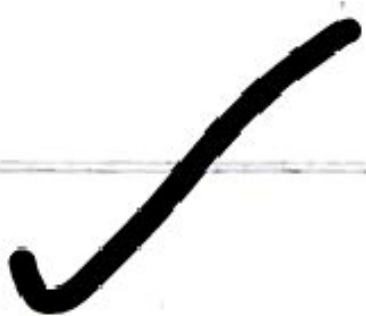
STATISTICS - 1

Question # 4 of 10 (Start time: 02:51:49 PM, 26 May 2021)

The point to the left of which the area is 70% is known as

Select the correct option

☐	70th value
☐	Quartile
☐	70/30 rule
☒	70th percentile



Question # 3 of 10 (Start time: 02:50:41 PM, 26 May 2021)

Coefficient of Correlation is independent of Change of Scale. Symbolically, this property is shown as:

page no 77

Select the correct option

☒	$\rho_{UV} = \rho_{XY}$
☐	$\rho_{U+V} = \rho_{XY}$
☐	$\rho_{YX} = \rho_{XY}$
☐	$\rho_{U-V} = \rho_{XY}$



Search the web and Windows



Consider the following formula.

$$\frac{E[(X - \mu_1)(X - \mu_2)]}{\sigma_1 \sigma_2}$$

Select the correct option

☐	Covariance
☒	Correlation
☐	Joint Mean
☐	Joint Variance

page no 76

x +

89d9-5ee7-4cf1-95d7-2d55ffd092a8

ail New Tab

If $p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x}$ then

$$p(1) = \frac{12}{256}$$

$$p(1) = \frac{54}{256}$$

$$p(1) = \frac{108}{256}$$

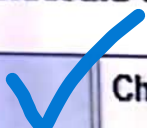


$$p(1) = \frac{81}{256}$$



The symbol $\varphi_z(t)$ is used to denote the

Select the correct option

☒

Characteristic Function

☐

Moment Generating Function

☐

Cumulative Distribution Function

☐

Probability Density Function

page no 46

Reload Math Equations

Question # 9 of 10 (Start time: 02:33:46 PM, 26 May 2021)

Total Marks: 1

Consider the following formula.

$$\frac{E[(X - \mu_1)(X - \mu_2)]}{\sigma_1 \sigma_2}$$

It can be used to calculate

Select the correct option

[Reload Math Equations](#)

☐	Covariance
☐	Joint Mean
☐	Joint Variance
☒	Correlation

Question # 8 of 10 (Start time: 02:32:32 PM, 26 May 2021)

Median is possible for which distribution?

Total Marks: 1

Select the correct option



Continuous distribution only



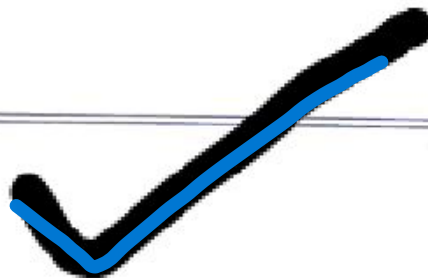
None of the given options



Both Discrete and Continuous



Discrete distribution only



Question # 7 of 10 (Start time: 02:31:33 PM, 26 May 2021)

The sum of the first n natural numbers

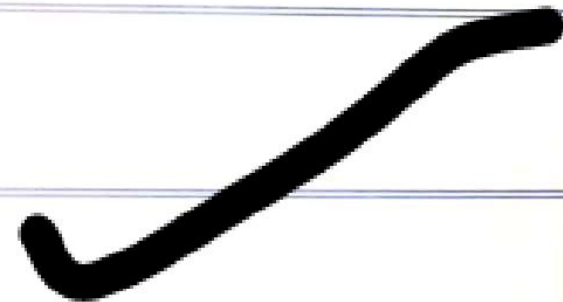
Total Marks: 1

Select the correct option

Reload Math Equations

☐	$n(n + 2)$
☐	$n(n + 1)$
☐	$\frac{n(n + 2)}{2}$
☒	$\frac{n(n + 1)}{2}$

page no 5



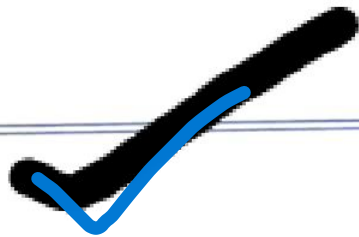
Click to view answers & Mark to Next Question

Question # 6 of 10 (**Start time: 02:30:31 PM, 26 May 2021**)

The probability of an impossible event is

Select the correct option

☐	infinity
☐	$1/2$
☐	1
☒	0



Question # 5 of 10 (Start time: 02:29:14 PM, 26 May 2021)

Total Marks: 1

If two variables X and Y are independent, coefficient of correlation between them will be

Select the correct option

☐	0
☐	1
☐	100
☐	2

page no 78

Question # 4 of 10 (Start time: 02:27:49 PM, 26 May 2021)

The joint probability distribution can be expressed in terms of a

Total M

Select the correct option

☐

joint probability mass function

☐

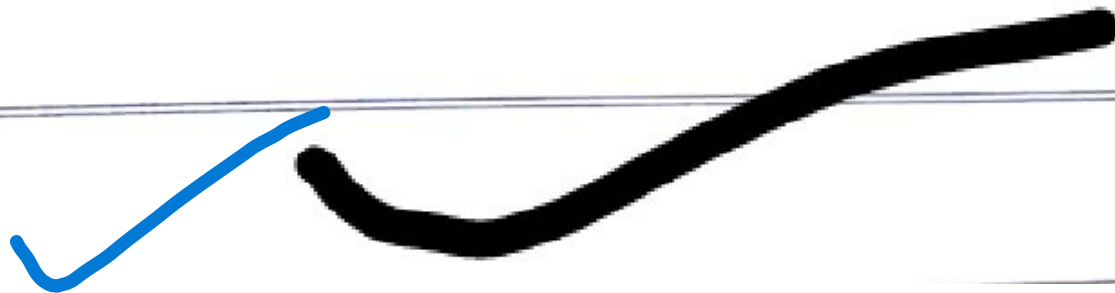
joint probability density function

☐

joint cumulative distribution function

☐

all of above three options



Question # 3 of 10 (Start time: 02:27:13 PM, 26 May 2021)

Total Marks: 1

What will be the value of

$$F_{XY}(\infty, \infty)$$

Select the correct option

☐	$F_{XY}(\infty, \infty) = 1$
☐	$F_{XY}(\infty, \infty) = 0$
☐	$F_{XY}(\infty, \infty) = \infty$
☐	$F_{XY}(\infty, \infty) = -\infty$

page no 58

Question # 2 of 10 (Start time: 02:26:27 PM, 26 May 2021)

Total Marks:

Inverse cumulative distribution function is also known as

Select the correct option

☒	quantile function
☐	multiplication function
☐	addition function
☐	probability function

02:25:21 PM, 26 May 2021)

The point to the left of which the area is 20% is known as

Select the correct option

☐	20th value
☐	80/20 rule
☐	Quartile
☐	20th percentile





MUHAMMAD SOHAIL

Time L

Quiz Start Time: 02:08 PM

(Start time: 02:23:01 PM, 26 May 2021)

In the continuous case of joint pdf $f_{X_1, X_2}(x_1, x_2)$, the marginal pdf of X_2 is found by integrating

ption

Reload M

Neither X_1 nor X_2 X_2 X_1 Both X_1 and X_2

Click to Save Answer & Move to Next Ques





, 26 May 2021)

For all pdfs, the value of $F_x(\infty)$

page no 18

1



- ∞

+ ∞

0





Question # 8 of 10 (Start time: 02:20:09 PM, 26 May 2021)

$F(x)$ is often called simply the

page no 11

Select the correct option



frequency function



negative function



distribution function



probability function





Question # 7 of 10 (Start time: 02:19:05 PM, 26 May 2021)

Toss a fair die; the possible values are 1, 2, 3, 4, 5, 6. Then probability is

Select the correct option

☐	$P(X = x) = \frac{3}{6}$
☐	$P(X = x) = \frac{1}{1 + 2 + 3 + 4 + 5 + 6}$
☐	$P(X = x) = \frac{1}{6}$
☐	$P(X = x) = \frac{6}{6}$



00203588: MUHAMMAD SOHAIL

42: QUIZ - 1

Question # 6 of 10 (Start time: 02:17:14 PM, 26 May 2021)

The expected value of e^{itX} is called

page no 46

Select the correct option

☒	Characteristic Function
☐	Cumulative Distribution Function
☐	Probability Density Function
☐	Moment Generating Function



MC200203588: MUHAMMAD SOHAIL

STA642:QUIZ - 1

Question # 5 of 10 (Start time: 02:16:09 PM, 26 May 2021)

Central moment generating function is denoted by

Select the correct option

☒ C(t)☐ CM☐ C☐ CentMom

page no 45

Click to S





FAU42. QUIZ - 1

Question # 4 of 10 (Start time: 02:14:10 PM, 26 May 2021)

What will be the value of

$$F_{XY}(\infty, \infty)$$

Select the correct option

☐	$F_{XY}(\infty, \infty) = 0$
☐	$F_{XY}(\infty, \infty) = \infty$
☒	$F_{XY}(\infty, \infty) = 1$
☐	$F_{XY}(\infty, \infty) = -\infty$



Question # 3 of 10 (Start time: 02:13:01 PM, 26 May 2021)

The concept of the CDF, will always be _____

Select the correct option

- | | |
|----------------------------------|-----------------------------------|
| ☐ | monotonically decreasing function |
| ☒ | monotonically increasing function |
| ☐ | monotonically zig-zag function |
| ☐ | none of these |





:200203588: MUHAMMAD SOHAIL

1642:QUIZ - 1

Quiz St

Question # 2 of 10 (Start time: 02:11:13 PM, 26 May 2021)

For a function $x = \frac{1}{2\sqrt{y}}$, what will be the value of x for y = 0.2

Select the correct option

☐	0.2
☐	0.50
☐	0.79
☒	1.12

Click To Save Answer





The Cumulative distribution graph for the discrete case is like a

Select the correct option

☐ straight line

☐ spiral

☐ zig-zag

☒ staircase

Click to Save Answer & Move to Next Question



MC200201157: MUHAMMAD ANSAR IQBAL

Time Left 85 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 05:04 PM, 26 May 2021

Question # 10 of 10 (Start time: 05:08:34 PM, 26 May 2021)

Total Marks: 1

Let X be a random variable with cumulative distribution function $F(x)$.
If $\lim_{x \rightarrow -\infty} F(x)$ then lower limit of $F(x)$ is

Select the correct option

Reload Math Equations

☐	$F(x) = 1$
☐	$F(x) = -1$
☐	$F(x) = -\infty$
☒	$F(x) = 0$



Click to Save Answer & Move to Next Question

page no 14



MC200201157: MUHAMMAD ANSAR IQBAL

Time Left 49 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 05:04 PM, 26 May 2021

Question # 9 of 10 (Start time: 05:07:48 PM, 26 May 2021)

Total Marks:

Let $X = (X_1, X_2)$ be a random vector.
Then the expected value of X exists if the expectation(s) of

Select the correct option

[Reload Math Equations](#)

☐	X_1 and X_2 does not matter
☒	X_1 and X_2 both exist
☐	X_1 exist only
☐	X_2 exist only

[Click to Save Answer & Move to Next Question](#)



Question # 8 of 10 (Start time: 05:06:51 PM, 26 May 2021)

For the joint PMF $p_{Y_1, Y_2}(y_1, y_2)$, we may obtain the Marginal PMF of y_1 by summing on ____

Select the correct option

☐	y_1
☒	y_2
☐	Neither y_1 nor y_2
☐	Either y_1 or y_2



Click to Save Answer &





MC200201157: MUHAMMAD ANSAR IQBAL

STA642-QUIZ - 1

Time Left 67 sec(s)

Quiz Start Time: 05:04 PM, 26 May 2021

Question # 7 of 10 (Start time: 05:06:24 PM, 26 May 2021)

Total Marks:

The probability of a sure event is

Select the correct option

- ☒ 1
- ☐ 1/2
- ☐ infinity
- ☐ 0

[Click to Save Answer & Move to Next Question](#)



MC200201157: MUHAMMAD ANSAR IQBAL

STA642-QUIZ - 1

Time Left 84 sec(s)

Quiz Start Time: 05:04 PM, 20 May 2021

Question # 6 of 10 (Start time: 05:06:14 PM, 20 May 2021)

Total Marks:

Theoretically there can be _____ of decimal places after the decimals.

Select the correct option

- ☒ an infinite number ✓
- ☐ finite number
- ☐ two number
- ☐ fixed number

Click to Save Answer & Move to Next Question

page no 6



The correlation coefficient lies between -1 and +1. It can be written as

Select the correct option

- ☒ $|p| < 1$
- ☐ $|p| < 0$
- ☐ $|p| > 1$
- ☐ $|p| > -1$

Click to Save Answer & Move to Next Question



MC200201157: MUHAMMAD ANSAR IQBAL

STA642:QUIZ - 1

Quiz Start

Question # 1 of 10 (Start time: 05:04:38 PM, 26 May 2021)

$$E(X_2) = E[E(X_2|X_1)]$$

The above formula is sometimes known as :

Select the correct option

☐	law of second expectation
☐	law of variance
☒	law of iterated expectations
☐	law of double expectation

Click to Save Answer

page no
75



MC200201157: MUHAMMAD ANSAR IQBAL

Time Left 85 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 05:04 PM, 26 May 2021

Question # 2 of 10 (Start time: 05:04:50 PM, 26 May 2021)

Total Marks:

The point to the left of which the area is 20% is known as

Select the correct option

- | | |
|----------------------------------|-----------------|
| ☐ | Quartile |
| ☒ | 20th percentile |
| ☐ | 20th value |
| ☐ | 80/20 rule |

Click to Save Answer & Move to Next Question



MC200201157: MUHAMMAD ANSAR IQBAL

Time Left 68 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 05:04 PM, 26 May 2021

Question # 3 of 10 (Start time: 05:04:57 PM, 26 May 2021)

Total Marks: -

Joint probability distribution of two variables is called

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☐ | Normal Distribution |
| ☐ | Multivariate distribution |
| ☐ | Binomial distribution |
| ☒ | Bivariate distribution |

Click to Save Answer & Move to Next Question



MC200200733: LUBNA NAHEED

Time Left 84 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 10 of 10 (Start time: 04:59:20 PM, 26 May 2021)

Total Marks: 1

The symbol $\varphi_x(t)$ is used to denote the

Select the correct option

Reload Math Equations

☐	Moment Generating Function
☐	Probability Density Function
☐	Cumulative Distribution Function
☒	Characteristic Function



Click to Save Answer & Move to Next Question



MC200200733: LUBNA NAHEED

Time Left 58 sec(s)

Quiz Start Time: 04:55 PM, 26 May 2021

STA642-QUIZ - 1

Total Marks: 1

Question # 7 of 10 (Start time: 04:57:41 PM, 26 May 2021)

Generally, Probability density function is symbolically denoted by

Select the correct option

☐	$f(x)$
☐	$d(x)$
☒	$F(x)$
☐	$L(x)$

Click to Save Answer & Move to Next Question



MC200200733: LUBNA NAHEED

STA642-QUIZ - 1

Time Left 64 sec(s)

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 8 of 10 (Start time: 04:58:16 PM, 26 May 2021)

Total Marks: 1

Let X be a random variable with cumulative distribution function $F(x)$.
If $\lim_{x \rightarrow \infty} F(x) = 1$ then upper limit of $F(x)$ is

Select the correct option

page no 14

☐	$F(x) = 0$
☒	$F(x) = 1$
☐	$F(x) = -\infty$
☐	$F(x) = -1$

Reload Math Equations

Click to Save Answer & Move to Next Question



MC200200733: LUBNA NAHEED

STA642:QUIZ - 1

Time Left 61 sec(s)

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 9 of 10 (Start time: 04:58:47 PM, 26 May 2021)

Total Marks:

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

Select the correct option

Reload Math Equations

- | | |
|----------------------------------|--------------------------|
| ☐ | $p(1) = \frac{54}{256}$ |
| ☒ | $p(1) = \frac{81}{256}$ |
| ☐ | $p(1) = \frac{12}{256}$ |
| ☐ | $p(1) = \frac{108}{256}$ |



Click to Save Answer & Move to Next Question



MC200200733: LUBNA NAHEED

Time Left 63 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 6 of 10 (Start time: 04:57:11 PM, 26 May 2021)

Total Marks: 1

In a continuous case, what will be the value of CDF for all x values which are less than zero?

Select the correct option

- | | |
|----------------------------------|-------------|
| ☒ | $F(x) = 0$ |
| ☐ | $F(x) = x$ |
| ☐ | $F(x) = -1$ |
| ☐ | $F(x) = 1$ |

page no 12

Click to Save Answer & Move to Next Question





MC200200733: LUBNA NAHEED

Time Left 77 sec(s)

STA642-QUIZ - 1

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 3 of 10 (Start time: 04:56:00 PM, 26 May 2021)

Total Marks: *

In probability, an event is known as _____ .

page no 2

Select the correct option

- ☐ a single value of sample space
- ☐ only even/odd values of sample space
- ☒ any subset of sample space
- ☐ all values of sample space

[Click to Save Answer & Move to Next Question](#)

Select the correct option

☐	$\rho_{U \cdot V} = \rho_{XY}$
☐	 $\rho_{UV} = \rho_{XY}$
☐	$\rho_{U - V} = \rho_{XY}$
☒	$\rho_{YX} = \rho_{XY}$

page no 77

CL

Question # 10 of 10 (Start time: 04:51:46 PM, 26 May 2021)

Total Marks: 1

The quantile function is also called the

Select the correct option

☐	cumulative distribution function
☒	inverse cumulative distribution function
☐	mod function
☐	median function

Question # 7 of 10 (Start time: 04:48:39 PM, 26 May 2021)

Total Marks: 1

In one-to-one transformation, for two old variables, we get

Select the correct option

☐	two new variables
☐	four new variables
☐	three new variables
☒	one new variable



MC200200733: LUBNA NAHEED

Time Left 64 sec(s)

STA642: QUIZ - 1

Quiz Start Time: 04:55 PM, 26 May 2021

Question # 1 of 10 (Start time: 04:55:09 PM, 26 May 2021)

Total Marks: 1

Any subset of a sample space is called:

Select the correct option

☐	exponential result
☐	Outcome
☒	event
☐	result

Click to Save Answer & Move to Next Question



Question # 5 of 10 (Start time: 04:47:12 PM, 26 May 2021)

Total Marks: 1

The point to the left of which the area is 35% is known as

Select the correct option

☒	35th percentile
☐	Quartile
☐	65/35 rule
☐	35th value

Question # 4 of 10 (Start time: 04:45:55 PM, 26 May 2021)

Total Marks: 1

Covariance is denoted by

Select the correct option

☐	$C(X,Y)$
☐	CVR
☒	$Cov(X,Y)$
☐	CV

Question # 3 of 10 (Start time: 04:45:05 PM, 26 May 2021)

Total Marks: 1

If random variable X is infinite then its expectation

Select the correct option

☐	exist
☐	minimum
☒	does not exist
☐	is finite

Question # 8 of 10 (Start time: 04:38:38 PM, 26 May 2021)

Total Marks: 1

$K(t)$ is equal to

Select the correct option

☐ $M(t)$ ☐ $E(M(t))$ ☒ $\log M(t)$ ☐ $\text{Var}(M(t))$

Question # 9 of 10 (Start time: 04:39:23 PM, 26 May 2021)

Total Marks: 1

$$\text{If } p(x) = \left(\frac{4!}{x!(4-x)!} \right) \left(\frac{1}{4} \right)^x \left(\frac{3}{4} \right)^{4-x} \text{ then}$$

Select the correct option

[Reload Math Equations](#)

☐	$p(2) = \frac{12}{256}$
☒	$p(2) = \frac{54}{256}$
☐	$p(2) = \frac{108}{256}$
☐	$p(2) = \frac{81}{256}$

[Go to Question / Previous Question / Next Question](#)

Question # 10 of 10 (Start time: 04:41:09 PM, 26 May 2021)

Total Marks: 1

The joint probability distribution can be expressed in terms of a

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | joint cumulative distribution function |
| ☐ | joint probability mass function |
| ☒ | all of above three options |
| ☐ | joint probability density function |

[Click to Save Answer](#) | [Change Next Question](#)

For any distribution, if a higher moment exists, then

_____.

Select the correct option

- | | |
|-----------------------|--|
| ☐ | very next higher moments necessarily exists. |
| ☐ | all higher moments necessarily exists. |
| ☐ | all lower moments may not exists. |
| ☐ | all lower moments necessarily exists. |



Click to Give Feedback / Move to Next Question

Question # 6 of 10 (Start time: 04:36:15 PM, 26 May 2021)

Total

For all pdfs, the value of $F_x(\infty)$

Select the correct option

[Reload Math E](#)

- | | |
|----------------------------------|-----------|
| ☐ | $-\infty$ |
| ☒ | 1 |
| ☐ | 0 |
| ☐ | $+\infty$ |

[Click to Save / Answer / Move to Next Question](#)

Question # 4 of 10 (Start time: 04:34:33 PM, 26 May 2021)

Total Marks: 1

If we subtract or add any constant in ALL values of X and Y, it will not affect the coefficient of correlation.
This property is known as:

Select the correct option

- | | |
|---|---|
| ☒ | Coefficients of Correlation are independent of Change of Origin |
| ☐ | Coefficients of Correlation are dependent of Change of Scale |
| ☐ | Coefficients of Correlation are independent of Change of Scale |
| ☐ | Coefficients of Correlation are dependent of Change of Origin |

[Click to Start / Answer / Move to Next Question](#)

Question # 3 of 10 (Start time: 04:33:18 PM, 26 May 2021)

Total Ma

If x and y are two real numbers then what is the relationship between the absolute value (the modulus) of $|x+y|$ and the modulus of $|x|$ and modulus of $|y|$?

Select the correct option



☒	$ x + y \leq x + y $
☐	$ x + y \neq x + y $
☐	$ x + y \geq x + y $
☐	$ x + y = x + y $

[Go to Question 2](#) / [Go to Next Question](#)

Question # 2 of 10 (Start time: 04:32:08 PM, 26 May 2021)

Total Marks: 1

If two variables X and Y are independent, coefficient of correlation between them will be

Select the correct option

☒	0
☐	2
☐	100
☐	1

[Click to Save Answer & Move to Next Question](#)

Let X be a random variable with cumulative distribution function $F(x)$.

If $\lim_{x \rightarrow -\infty} F(x)$ then lower limit of $F(x)$ is

Select the correct option

 Reload Math Equations

☐

$$F(x) = 1$$

☐

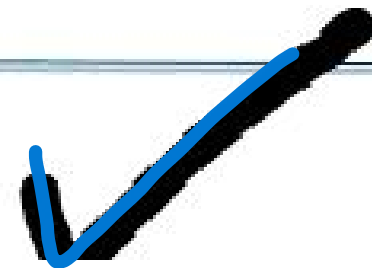
$$F(x) = -\infty$$

☐

$$F(x) = -1$$

☐

$$F(x) = 0$$



Question # 8 of 10 (Start time: 04:26:46 PM, 26 May 2021)

Central moment generating function is denoted by

Select the correct option

☐ C☐ CentMom☐ CM☒ $C(t)$ [Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 04:25:42 PM, 26 May 2021)

Total Marks:

Theoretically there can be _____ of decimal places after the decimals.

Select the correct option



finite number



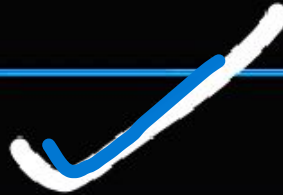
two number



fixed number



an infinite number

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 04:19:31 PM, 26 May 2021)

In Probability, CDF stands for

Select the correct option

- | | |
|----------------------------------|----------------------------------|
| ☐ | Central Distribution Function |
| ☐ | Complete Distribution Function |
| ☐ | Cumulative Deviation Function |
| ☒ | Cumulative Distribution Function |

[Click to Save Answer & Move to Next Question](#)

Question # 6 of 10 (Start time: 04:23:37 PM, 26 May 2021)

Total Marks:

In a continuous case, what will be the value of CDF for all those x values that lies between 0 and 1?

Select the correct option

☐ $F(x) = 0$

☐ $F(x) = 1$

☒ $F(x) = x$

☐ $F(x) = 0.5$

[Click to Save Answer & Move to Next Question](#)

Question 6 of 10 (Start time: 04:17:12 PM, 20 May 2021)

Total Marks

As the random variables, X_1 and X_2 are continuous random variables, the difference of these random variables is

Select the correct option

- ☐ discrete random variable
- ☒ continuous random variable
- ☐ both continuous and discrete
- ☐ not existent

[Click to Move Answer & Move to Next Question](#)



MC200200242: AMNA ANWAR

Time Left 84 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 04:12 PM, 26 May 2021

Question # 1 of 10 (Start time: 04:12:32 PM, 26 May 2021)

Total Marks: 1

For all pdfs, the value of $F_p(\infty)$

Select the correct option

Reload Math Equations

☐	$+\infty$
☐	0
☐	$-\infty$
☒	1

Click to Save Answer & Move to Next Question





MC200200829: MUHAMMAD TARIQ

Time Left 81 sec(s)

STA042: QUIZ - 1

Quiz Start Time: 03:39 PM, 28 May 2021

Question # 8 of 10 (Start time: 04:02:20 PM, 28 May 2021)

Total Marks:

If we subtract or add any constant in ALL values of X and Y, it will not affect the coefficient of correlation. This property is known as:

Select the correct option

- ☐ Coefficients of Correlation are independent of Change of Scale
- ☐ Coefficients of Correlation are dependent of Change of Origin
- ☒ Coefficients of Correlation are independent of Change of Origin
- ☐ Coefficients of Correlation are dependent of Change of Scale

Previous Question Next Question





The Cumulative distribution graph for the continuous case is like a

Select the correct option

- ☐ spiral
- ☐ zig-zag
- ☐ staircase
- ☒ curve or straight line



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Joint probability distribution of two variables is called

Select the correct option

- ☐ Binomial distribution
- ☐ Multivariate distribution
- ☒ Bivariate distribution
- ☐ Normal Distribution

Correct Answer: Bivariate distribution





If random variable X is infinite then its expectation

Select the correct option

- ☐ minimum
- ☐ exist
- ☒ does not exist
- ☐ is finite

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In probability, PDF stands for

Select the correct option

- ☒ Probability Density Function
- ☐ Probability Deviation Function
- ☐ Probability Distribution Function
- ☐ Probability Discrete Function

Correct Answer: Probability Density Function





MC200200839: MUHAMMAD TARIQ

Time Left: 80 min(x)

STA042 QUIZ - 1

Quiz Start Time: 03:34 PM, 20 May 2022

Question # 1 of 10 (Start time: 04:50:34 PM, 20 May 2021)

Total Marks:

Let $X = (X_1, X_2)$ be a random vector.
Then the expected value of X exists if the expectation(s) of

Select the correct option

Related Math Equations

☐	X_1 and X_2 both exist
☐	X_2 exist only
☐	X_1 and X_2 does not matter
☐	X_1 exist only



Get the answers of your questions



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1



MC200200829: MUHAMMAD TARIQ

Time Left 87 sec(s)

STA642:QUIZ - 1

Quiz Start Time: 03:54 PM, 28 May 2021

Question # 10 of 10 (Start time: 04:08:27 PM, 28 May 2021)

Total Marks:

What is the name of the following series?

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Select the correct option

Related Math Equations

☒	Maclaurin series
☐	Exponential series
☐	Binomial series
☐	Taylor series

Previous Question Next Question

Quiz 1

Theoretically there can be an infinite number of decimal places after the decimals. Page 62

Quiz 2 . Coefficients of Correlation are independent of Change of Origin:

This property reveals that if we subtract any constant from all the values of X (or add any constant to all the values of X) Page 865

. Coefficient of Correlation is independent of Change of Scale:

This property reveals that if we divide or multiply all the values of X by any constant and divide or multiply all the values of Y by any constant, it will not affect the coefficient of correlation

Quiz 3 Jacobian (denoted by J) of the transformation. Page 254

Quiz 4 say, 20% is known as the 20th percentile, the point to the left of which the area is, say, 35% is known as the 35th percentile and so Page 296

Quiz 5 • The quantile function is also called the inverse cumulative distribution function Page 311

Quiz 6 The joint probability distribution can be expressed either in terms of a joint cumulative distribution function or in terms of a joint probability density function (in the case of continuous variables) or joint probability mass function (in the case of discrete variables). Page 583

Quiz 7 (Cumulative Distribution Function) the discrete case it is like a staircase Page 114

Quiz 8 Coefficient of Correlation is independent of Change of Scale: $P_{UV} = P_{XY}$ Page 867

Quiz 9 Let X be a random vector. Then the expected value of X exists if the expectations of X_1 and X_2 exist. Page 738

Quiz 10 The central moment generating function is given denoted by $C(t)$ Page 515

Quiz 11 correlation
$$\rho = \frac{[(x - u_1)(x - u_2)]}{\sigma_1 \sigma_2} = \frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}$$
 Page 850

Quiz 12 6. If two variables X and Y are independent, coefficient of correlation between them will be zero. Page 873

Quiz 13 Any experiment or process in which the outcome is unpredictable that is eligible to be called a random Page 3

Quiz 14 For any distribution, if a higher moment exists, then all lower moments necessarily exists. Page 815

Quiz 15 The conditional probability of even A, given event B, as

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = P(B) \neq 0 \text{ Page 819 This number is called the covariance of X and Y and}$$

is often denoted by $\text{cov}(X,Y)$. Page 849

Quiz 16 If x and y are two real numbers then the absolute value (the modulus) of $|x+y|$ is less than or equal to the modulus of $|x|$ and modulus of $|y|$ $\|x+y\| \leq \|x\| + \|y\|$ Page 752

Quiz 17 Let X be a random variable with cumulative distribution function $F(x)$.

Then, (b) $\lim_{x \rightarrow -\infty} F(x) = 0$ (the lower limit of F is 0). Page 152

Quiz 18 cdf the joint cumulative distribution function Page 587

Quiz 19 Maclaurin's series Page 447

Quiz 20 probability density function (pdf) Page 64

Quiz 21 a random variable f it is finite, then the expectation exists and we can determine its expectation by obtaining the distribution and if it is infinite (i.e. if it is not finite), we shall say that the expectation of X does not exist Page 688 2 Quiz

Quiz 22 joint probability distribution In the case of only two random variables, this is called a bivariate distribution Page 582

Quiz 23 For a continuous random variable X , the CDF is as follows:

- $F(x) = 0$ for all x values which are less than zero
- $F(x) = x$ for all those x values that lies between 0 and 1.
- $F(x) = 1$ for all those x values are greater than 1. Page 130

$$\text{Quiz 24 } E(X_2) = E[E(X_2 / X_1)]$$

This is sometimes called the law of total expectation and sometimes it is called the law of iterated expectations. Page 844

Quiz 25 Sample Space It is the set of all possible outcomes of a random experiment Page 8

Quiz 26 probability density function $f(x)$ Page 692

Quiz 27 The concept of the CDF, will always be monotonically increasing function. Page 194

Quiz 28 The Characteristic function e^{itx} $\varphi_X(t)$ Page 794 2 quiz

Quiz 29 $P(X=x)=1/6$ fair die 1,2,3,4,5,6 Page 43

Quiz 30 cumulative distribution function $F(x)$. Page 57

Quiz 31

- In the continuous case, the marginal pdf of X_1 is found by integrating out x_2 .



• Similarly, the marginal pdf of X_2 is found by integrating out x_1 . Page 668

Quiz 32 the value of $F_{xy}(\infty, \infty) = 1$ Page 637

Quiz 33 The Probability of an impossible event. Is 0 Page 153

Quiz 34 pdfs satisfy the two properties

1) $f_X(x) > 0$ and (ii) $\int_{-\infty}^{\infty} f_X(t) dt = 1$

2) The second property, of course, follows from $F_X(\infty) = 1$ Page 200

Quiz 35 The sum of the first n natural numbers $\frac{n(n+1)}{2}$ Page 55

Quiz 36 The median of a distribution of a random variable X of the discrete or continuous type is a value of x such that Page 282

Quiz 37 $x = \frac{1}{2\sqrt{y}}$, $y = 0.6$

$$x = \frac{1}{2\sqrt{0.6}} = \frac{1}{1.549193} = 0.6454 = 0.65$$

